AIoT-Based Posture and Fatigue Monitoring Using TinyML and Multimodal Sensors

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***Abstract*—This paper presents an AIoT-based system for real-time monitoring of posture and fatigue during desk work using embedded TinyML techniques. The proposed solution integrates computer vision and inertial sensing on a compact edge device based on the XIAO ESP32-S3 Sense microcontroller. Two lightweight convolutional neural networks based on MobileNetV2 are deployed to detect sitting posture and fatigue-related gestures from grayscale images captured by an onboard camera. In parallel, an MPU6050 accelerometer measures upper-body orientation to provide complementary posture information. A lightweight sensor fusion strategy combines vision-based predictions with inertial measurements to improve detection robustness. The system performs all inference locally on the device, achieving near real-time response with latency between 200 and 250 ms while maintaining low memory usage. Detected events are transmitted through MQTT to the Ubidots platform for remote monitoring and visualization. Experimental results demonstrate that the proposed architecture can successfully execute multiple neural network models on resource-constrained hardware while maintaining high classification accuracy and reliable real-time performance.**

I. Introduction

The increasing prevalence of computer-based work has led to a significant rise in sedentary behavior and prolonged sitting periods among office workers and students. Although digital technologies have improved productivity, they have also contributed to the development of musculoskeletal disorders associated with poor posture and fatigue during extended desk work [1]. Maintaining proper ergonomic posture is essential to prevent chronic back pain, neck strain, and reduced work performance [2].

Traditional ergonomic interventions typically rely on periodic supervision, manual posture training, or external devices such as pressure cushions and wearable sensors. However, these approaches often require specialized equipment, continuous human monitoring, or intrusive instrumentation, limiting their scalability and usability in everyday work environments.

Recent advances in Artificial Intelligence (AI), computer vision, and the Internet of Things (IoT) have enabled the development of intelligent systems capable of monitoring human posture automatically. In particular, lightweight deep learning models can now run directly on embedded devices through TinyML, enabling real-time inference at the edge without requiring cloud connectivity. This approach reduces latency, improves privacy, and allows continuous monitoring in resource-constrained devices.

Several studies have explored AI-based posture detection systems. Zhao and Su [3] proposed a computer vision method to recognize sitting posture using camera-based observation. Similarly, Sahoo et al. [4] introduced an AI-driven IoT framework for real-time posture monitoring in workplace environments. Other approaches rely on embedded sensors such as accelerometers or pressure sensors to detect body orientation and posture patterns [5]. While these systems demonstrate promising results, many of them require multiple sensors, external processing units, or continuous cloud connectivity.

To address these limitations, this work proposes an AIoT-based intelligent posture and fatigue monitoring system designed for desk-based work environments. The proposed system integrates computer vision and inertial sensing using a compact embedded platform based on the ESP32-S3 Sense microcontroller. Two lightweight deep learning models based on MobileNetV2 are deployed on-device to detect both sitting posture and fatigue-related gestures in real time. Additionally, an inertial measurement unit (IMU) attached to the user's upper back provides complementary orientation data, allowing a sensor fusion strategy that improves posture estimation reliability.

The main contributions of this work are summarized as follows:

- Design of a low-cost AIoT architecture for real-time ergonomic monitoring using an ESP32-S3 embedded platform.
- Implementation of two TinyML models for posture classification and fatigue gesture detection using MobileNetV2 with grayscale image preprocessing.
- Integration of computer vision and inertial sensing through a lightweight sensor fusion mechanism to improve detection robustness.
- Deployment of a real-time monitoring pipeline with edge inference and IoT event logging using the Ubidots platform.

The proposed system demonstrates the feasibility of implementing intelligent ergonomic monitoring entirely on embedded hardware, enabling real-time posture feedback while minimizing computational resources and network dependency.

II. Proposed Solution

The architecture of the system follows a three-layer AIoT paradigm consisting of: (i) a perception layer responsible for sensing and data acquisition, (ii) a computation layer where data processing and machine learning inference occur, and (iii) a communication layer that enables remote monitoring and data visualization.

Figure 1 presents the overall system architecture.

A. Perception Layer

The perception layer is responsible for acquiring the physical signals required for posture and fatigue analysis. Two sensing modalities are employed in the system:

- **Vision sensor:** An OV3660 camera integrated in the XIAO ESP32-S3 Sense captures images of the user during desk work. These images are used to detect body posture and fatigue-related gestures.
- **Inertial sensor:** An MPU6050 accelerometer provides inertial measurements of the user's upper back orientation. The sensor is physically attached to the upper back of the user, slightly below the collar area of the shirt, allowing the system to capture trunk inclination through pitch and roll angles.

The use of two sensing modalities enables a complementary representation of posture, where visual information captures body configuration while inertial data provides orientation measurements.

B. Computation Layer

The computation layer runs entirely on the XIAO ESP32-S3 microcontroller and is responsible for signal preprocessing, machine learning inference, sensor fusion, and local decision making.

First, captured images are preprocessed by converting them to grayscale and resizing them to a resolution of 96×96 pixels in order to reduce computational cost and memory usage. The processed images are then fed into two TinyML models deployed on the device using the Edge Impulse framework.

The first model performs fatigue gesture detection, identifying gestures such as yawning or eye rubbing. The second model performs posture classification, detecting four possible sitting states: good posture, forward head posture, slouched posture, and lateral inclination.

In parallel, accelerometer readings are processed to estimate body orientation. During an initial calibration stage, the system records the user's neutral posture and stores the corresponding pitch and roll reference values. These reference values are later used to compute deviations from the normal posture.

Thresholds used for posture deviation detection were empirically calibrated during preliminary experiments in order to distinguish between natural small movements and sustained incorrect postures.

The outputs of the vision-based posture classifier and the inertial classifier are combined through a lightweight sensor fusion mechanism. The final posture estimation is obtained using a weighted combination where the computer vision model contributes the majority of the decision while the inertial sensor provides a corrective component. This strategy improves robustness against lighting variations, camera occlusions, or temporary tracking errors.

Finally, a temporal activation logic evaluates whether the detected state persists for a predefined duration before triggering a feedback signal. This timer-based validation prevents false alarms caused by momentary posture adjustments or short transient movements that naturally occur during desk work.

C. Communication Layer

The communication layer enables remote monitoring of the system through IoT connectivity. Once a posture anomaly or fatigue gesture is confirmed by the fusion and temporal validation logic, the embedded system generates a JSON message containing the current state of the user and publishes it through an MQTT client.

The data is transmitted over WiFi to the Ubidots cloud platform, where it is stored and visualized through a real-time dashboard. This interface allows users to observe the current posture and fatigue status, along with historical counters specifically dedicated to tracking incorrect postures and fatigue-related gestures during work sessions. Figure 2 shows Ubidots setup.

This architecture enables both real-time local feedback via physical LEDs and long-term ergonomic monitoring through cloud-based data aggregation.

Fig. 2. Ubidots dashboard used for real-time visualization of posture and fatigue monitoring data.

D. Physical Implementation

The physical deployment of the system consists of a single embedded node based on the XIAO ESP32-S3 Sense microcontroller. The device is positioned facing the user on the desk in order to capture upper body posture through the camera. The MPU6050 sensor is attached to the user's upper back to measure trunk inclination.

Figure 3 illustrates the physical setup used in the experimental evaluation.

E. Posture Probability and Sensor Fusion

The accelerometer provides information about the orientation of the user's body through the pitch and roll angles. These angles are computed from the acceleration components measured by the MPU6050 sensor.

Let (a_x, a_y, a_z) denote the acceleration values measured along the three axes. The orientation angles are estimated as:

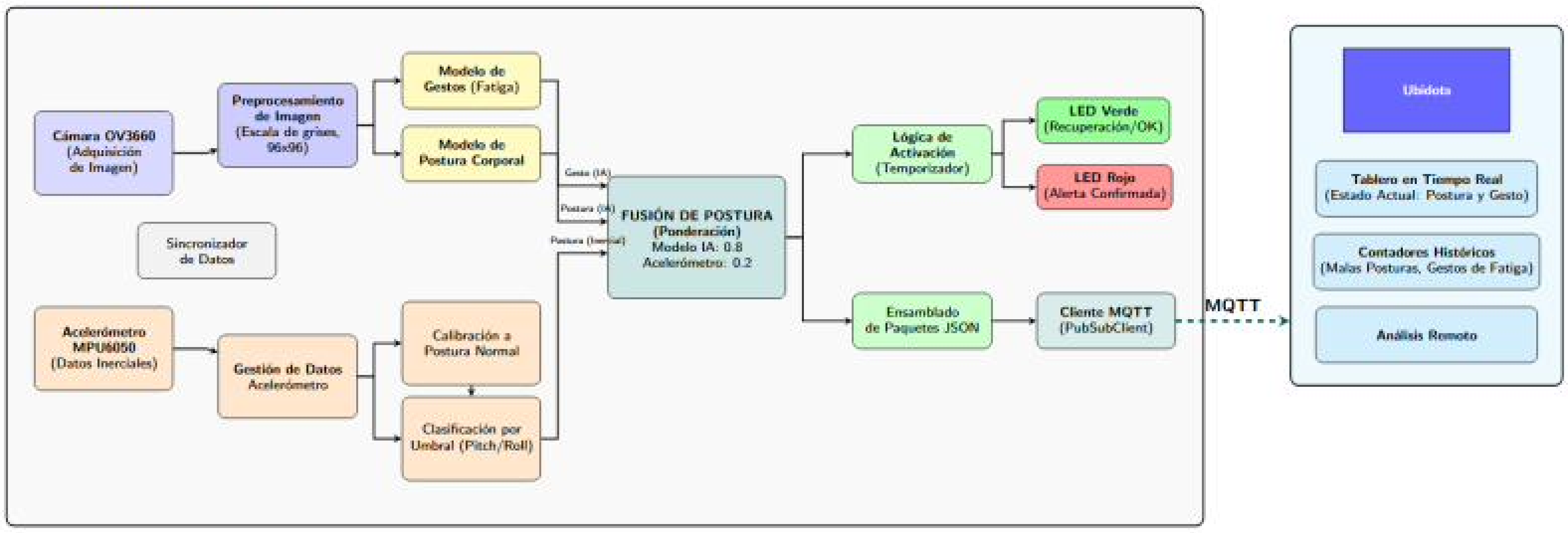

Fig. 1. AIoT system architecture for posture and fatigue monitoring implemented on the ESP32-S3 sensor node.

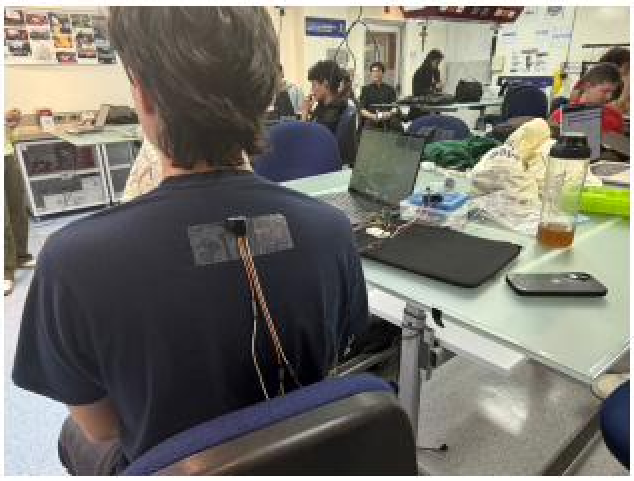

Fig. 3. Physical implementation of the proposed posture monitoring system.

$$\text{pitch} = \arctan\left(\frac{a_y}{\sqrt{a_x^2 + a_z^2}}\right), \qquad \text{roll} = \arctan\left(\frac{-a_x}{a_z}\right)$$

The deviation of the user from a calibrated neutral posture is measured using the absolute deviations of pitch and roll. A combined deviation magnitude is computed as:

$$D = \sqrt{\theta_p^2 + \theta_r^2}$$

where θ_p and θ_r represent the pitch and roll deviations relative to the neutral posture.

This deviation is normalized using a predefined threshold T in order to obtain a score interpreted as the probability of incorrect posture derived from the inertial sensor:

$$P_{acc} = \min\left(1, \frac{D}{T}\right)$$

The threshold T was experimentally calibrated during preliminary testing sessions in order to balance sensitivity and robustness, ensuring that normal posture adjustments do not immediately trigger alerts.

In parallel, the computer vision model running on the edge device predicts the probability of incorrect posture from the camera image, denoted as P_{IA}.

Both signals are combined using a weighted fusion approach:

$$S = w_{IA}P_{IA} + w_{ACC}P_{ACC}$$

where S is the final posture score. In the proposed system, the vision model is considered the main estimator, while the accelerometer provides complementary information. Therefore, the weights used are:

$$w_{IA} = 0.8, \qquad w_{ACC} = 0.2$$

An alert is triggered when the fused score S exceeds a predefined threshold and remains above this value for a minimum time interval (3000 ms). This temporal validation mechanism prevents false alarms caused by short posture transitions or temporary body movements.

III. Implementation Details

A. Dataset Construction

The datasets were collected using the OV3660 camera integrated into the XIAO ESP32-S3 Sense module under realistic desk-working conditions. Image acquisition was performed directly from the device using the SenseCraft platform.

Two participants took part in the data collection process, both male, with approximate heights of 175 cm and 168 cm.

For the posture classification model, images were labeled into four categories:

- Good posture
- Forward head posture
- Slouched posture
- Lateral inclination

A total of 1623 images were collected from both participants in two different indoor environments to introduce background and lighting variability.

For the fatigue gesture detection model, three classes were defined:

- Normal state
- Eye rubbing
- Stretching-yawn gesture

This dataset contained 633 images and was collected using a single participant with slight background variations.

Initially, yawning alone was considered as a fatigue indicator. However, early experiments showed that the subtle visual changes involved made it difficult to detect reliably with the embedded camera. To improve robustness, the class was redefined as *stretching-yawn*, which produces stronger visual patterns.

Following the Edge Impulse pipeline, datasets were split into 80% training and 20% testing, with 20% of the training samples used for validation during training. This configuration allowed monitoring the generalization performance of the models while preserving an independent test set for final evaluation.

B. Image Preprocessing

All images were preprocessed before training to reduce computational complexity and adapt them to the constraints of the embedded platform.

The preprocessing pipeline included conversion from RGB to grayscale, resizing images to 96×96 pixels, and pixel normalization. During training, data augmentation was also enabled to introduce small variations such as rotations and translations. This proved particularly useful to improve the robustness of the models when deployed on the embedded device.

Feature extraction is performed automatically by the convolutional layers of the neural network. These layers learn relevant spatial patterns directly from the images, such as head position, shoulder alignment, and arm movements associated with fatigue gestures.

C. Neural Network Architecture

Both models use lightweight convolutional neural networks based on MobileNetV2, which are well suited for TinyML applications due to their low computational cost and reduced memory footprint.

The final architecture and hyperparameters were obtained through an iterative testing process. Different configurations were evaluated by varying parameters such as the number of neurons in the dense layers and observing both classification performance and execution behavior on the ESP32-S3 device.

For the posture classification model, a dense layer with 32 neurons provided the best balance between classification accuracy and computational cost. Smaller configurations were tested but showed reduced ability to capture the spatial patterns associated with different posture deviations.

For the fatigue gesture model, a smaller dense layer with 8 neurons was selected. Fatigue gesture detection is not the primary functionality of the system and experiments showed that 8 neurons were sufficient to achieve excellent performance while keeping the model lightweight.

TABLE I

NEURAL NETWORK ARCHITECTURE FOR FATIGUE GESTURE DETECTION

Input layer	96×96 grayscale image (9,216 features)
Backbone network	MobileNetV2 (width multiplier 0.35)
Dense layer	8 neurons
Dropout	0.10
Output layer	3 classes (stretching-yawn, normal, eyes)
Training epochs	20
Learning rate	0.0005
Batch size	32
Validation split	20%

TABLE II

NEURAL NETWORK ARCHITECTURE FOR POSTURE CLASSIFICATION

Input layer	96×96 grayscale image (9,216 features)
Backbone network	MobileNetV2 (width multiplier 0.35)
Dense layer	32 neurons
Dropout	0.15
Output layer	4 classes (good, forward head, slouched, lateral)
Training epochs	20
Learning rate	0.0005
Batch size	32
Validation split	20%

D. Training Results

The posture classification model achieved an accuracy of 99.2% on the validation set with a loss value of 0.03. The confusion matrix shows near-perfect classification across all posture categories. The weighted F1 score reached 0.99, with an area under the ROC curve of 1.00.

The fatigue gesture detection model achieved 100% validation accuracy with a loss value of 0.04. All classes were correctly classified in the validation set, resulting in a weighted F1 score of 1.00 and an ROC AUC of 1.00.

E. Model Deployment and Multi-Model Integration

The trained models were exported from Edge Impulse as Arduino inference libraries for deployment on the XIAO ESP32-S3 microcontroller.

Although Edge Impulse provides the EON Compiler for optimized embedded inference, several compatibility issues were encountered when attempting to integrate both models simultaneously using the EON optimized version. Due to these issues, the models were exported using the standard TensorFlow Lite for Microcontrollers implementation instead. While this approach results in slightly larger binaries, it provided a more stable integration when running both models on the same device.

Another implementation challenge was executing two Edge Impulse models within the same Arduino firmware. Each exported model contains similar internal definitions and inference functions, which can cause symbol conflicts during compilation.

To address this issue, a custom header file named `postura_impulse.h` was introduced. This file plays a key role in enabling multi-model inference on the embedded system.

The purpose of this header is to encapsulate the second model (PosturaFinal) and isolate its internal definitions so that it can coexist with the first model (Gestos) without causing naming conflicts or redundant code inclusion.

By using this structure, the firmware can sequentially execute inference for both neural networks within the same application, allowing the system to simultaneously perform fatigue gesture detection and posture classification on the edge device.

IV. Experimental Results, Analysis, and Limitations

A. Deployment Performance on Embedded Hardware

The final system was deployed on the XIAO ESP32-S3 Sense platform, executing both posture and gesture recognition models locally using the camera input. The measurements obtained directly from the embedded device indicate that the system operates comfortably within the hardware constraints.

The internal RAM usage was measured at 84,140 bytes out of 345,172 bytes (24.4%), while external PSRAM usage reached 30,372 bytes out of 8,385,767 bytes (0.4%). The compiled program occupied 2,175,200 bytes of flash memory out of the available 5,517,536 bytes (39.4%). These values demonstrate that the system maintains significant memory headroom, allowing the device to operate both models simultaneously.

Inference latency measured on-device showed that the Digital Signal Processing (DSP) stage required approximately 4 ms for both models. The classification stage required 199 ms for gesture detection and 222 ms for posture recognition. These results, shown in Figure 4 confirm that the system can perform inference in near real time, which is suitable for continuous ergonomic monitoring.

However, these measurements correspond to runtime observations rather than isolated model execution. For this reason, additional profiling provided by the Edge Impulse platform was also considered in order to better understand the computational characteristics of each model individually.

B. Model Resource Consumption

Edge Impulse provides detailed estimates of latency, RAM usage, and flash size for each trained model. Table III summarizes the main characteristics of both neural networks used in the system.

Both models exhibit similar RAM requirements due to the use of a comparable MobileNet-based transfer learning architecture. However, the posture model occupies slightly more flash memory than the gesture model.

Fig. 4. Example of real-time system output displayed in the Arduino serial monitor during posture and fatigue monitoring.

TABLE III
Resource usage of trained models (Edge Impulse profiling)

Model	Latency (Total)	RAM	Flash
Gesture Detection	1838 ms	353.1 KB	673.6 KB
Posture Detection	1645 ms	353.1 KB	704.6 KB

This difference is primarily attributed to architectural variations introduced during hyperparameter tuning. The posture detection network uses a larger number of neurons in the final dense layer with 32 neurons. In contrast, the gesture model uses only 8 neurons.

Despite these differences, both models remain compact enough to coexist within the flash memory of the embedded device.

C. Model Accuracy Evaluation

Model evaluation was performed using the testing metrics provided by Edge Impulse. Both networks achieved high performance, indicating strong generalization across the collected dataset.

The gesture detection model achieved an overall accuracy of 98.36%, with an Area Under the ROC Curve (AUC) of 1.00 and weighted precision, recall, and F1-score values of approximately 0.99. Most classifications were correctly identified, with only minor confusion between the "Stretch/Yawn" gesture and other states.

The posture detection model achieved an even higher accuracy of 99.41%, with similarly strong precision, recall, and F1-score metrics (all approximately 0.99). The confusion matrix shows that most posture classes are perfectly separable, with only small overlaps occurring between the "Good posture" and "Forward neck" or "Slouched" categories.

Overall, the results indicate that the trained models are capable of accurately identifying both ergonomic posture states and fatigue-related gestures.

D. Real-Time Testing

Beyond offline evaluation, the system was also tested using the SenseCraft platform, which allows real-time visualization

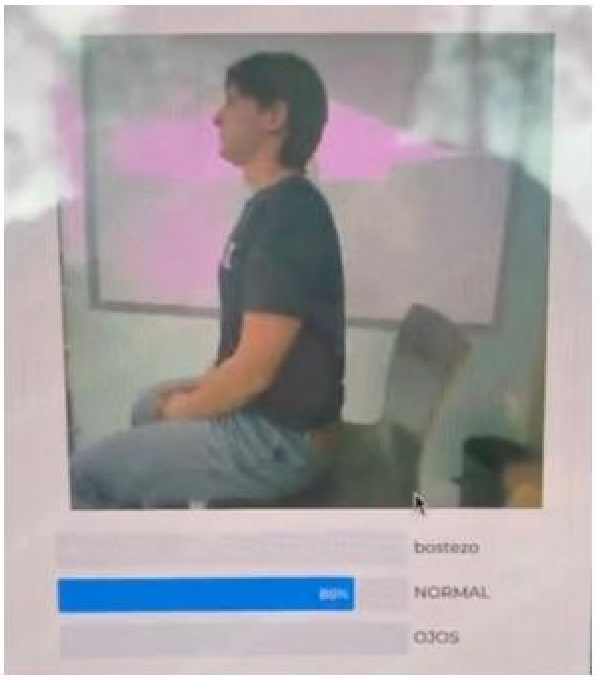

Fig. 5. Real-time model validation using the SenseCraft interface during the development phase

of the camera stream and inference results directly from the embedded device. An example of the visualization is illustrated in Figure 5.

This environment enabled interactive validation of the system by observing how the model predictions responded to live user movements. Real-time testing confirmed that the system can successfully identify posture deviations and fatigue gestures while the user is performing normal desk activities.

The ability to visualize predictions alongside the camera feed proved useful for verifying dataset quality, understanding occasional misclassifications, and validating that the models respond correctly under realistic operating conditions.

E. Limitations

Despite the strong experimental results, several practical limitations were identified during system testing.

First, the positioning of the camera relative to the user significantly affects detection quality. Improper camera angles or large variations in distance may reduce classification reliability, particularly for posture detection.

Second, variations in user body characteristics, such as height or body proportions, can influence the recognition of certain fatigue gestures. For example, gestures involving eye rubbing or stretching may appear slightly different depending on the user's physical build.

Finally, the system currently relies on a single camera perspective. While this simplifies deployment, it may limit robustness in environments where the user's position relative to the device changes frequently.

V. Conclusions

This work presented an AIoT-based system for real-time monitoring of posture and fatigue during desk work using an embedded TinyML architecture. The proposed solution integrates computer vision and inertial sensing on a single edge device to provide both local ergonomic feedback and remote monitoring capabilities.

Experimental results demonstrated the technical feasibility of deploying multiple neural network models on a resource-constrained microcontroller. The XIAO ESP32-S3 Sense was capable of executing two MobileNetV2-based models while simultaneously processing data from the MPU6050 sensor. The system achieved inference latencies between 200 and 250 ms, enabling near real-time detection of posture deviations and fatigue-related gestures.

The use of sensor fusion proved beneficial for improving posture detection reliability. While the vision-based model captures visual posture cues, the inertial sensor provides complementary information about upper-body orientation. Additionally, performing all inference locally on the device highlights the advantages of edge computing, allowing the system to operate with low latency and without requiring continuous internet connectivity.

VI. Future Work

Future improvements may focus on increasing the efficiency and scalability of the system. One important direction is energy optimization through event-driven activation, where the camera is only enabled when the accelerometer detects significant posture changes.

Hardware integration could also be improved by embedding inertial sensors into wearable textiles, ensuring stable sensor placement during daily use. Additionally, a multi-node architecture using multiple microcontrollers could separate vision and inertial processing, improving modularity and reducing wiring complexity.

Further enhancements may include the use of multiple inertial sensors to better differentiate neck and trunk movements, as well as expanding the training dataset with greater diversity in users, environments, and working conditions to improve model generalization.

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